

Monograph on
 α -designs

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PREAMBLE

It gives me a great pleasure to write a few lines about this monograph which is concerned with certain resolvable incomplete block designs, called Alpha (α -) designs. As the brief introduction below shows, such designs have a great utility. Indeed, here, the application has motivated this effort.

In designed experiments the two main sources of variability in the data generated are the *treatments* and the *experimental material*. Variability due to treatments arises because several treatments are tried in the experiment and it is expected that the treatment effects are different. This is a deliberate attempt on the part of the experimenter to generate variability. If all the treatments were expected to be same, then the experimenter would not run the experiment. However, the other major component of variability needs to be explained through the model. This can be explained by way of designing the experiment properly.

It has been observed that in many crop improvement programs conducted in the National Agricultural Research System (NARS), initial varietal trials and advanced varietal trials are generally conducted using a Randomized Complete Block (RCB) design. The analysis of the data generated reveals that the percent coefficient of variation (CV) in a large number of such experiments is high and consequently the precision of treatment comparisons is low. This amounts to saying that the error sum of squares is large compared to the sum of squares attributable to the model. Hence, small treatment differences may not be detected as significant.

Because of the above, in practice, a large number of trials get rejected because of high CV, resulting in a tremendous loss of resources and time. Generally in these experiments the block mean squares are small as compared to error mean squares meaning thereby that the blocking has not been effective in the sense that the blocks have not been formed properly. In field trials, the formation of blocks, size and shape of plots should be preceded by uniformity trials to determine the fertility pattern through a contour map. In most situations, particularly in the plains, the contour maps show the variations in the fertility in patches rather than a uniform pattern. Therefore, there is a need for effective control of the variation in the experimental material through proper blocking.

Incomplete block designs (like balanced incomplete block (BIB) design, partially balanced incomplete block (PBIB) design, cyclic design, etc.) are useful in such situations. However, one may argue that in these designs the demonstration of a varietal effect in the field cannot be done as all the treatments are not appearing in adjacent pieces of land. To overcome this problem, resolvable block designs with smaller block sizes may be used.

Resolvable block designs offer a great deal of flexibility in the sense that, by using them, experiments can be conducted with one replication at a time, with different replications applied at different places or at different points of time. For example, field trials with large number of crop varieties cannot always be laid out in a single location or in a single season. Therefore, it is desired that variation due to location or time periods may also be controlled along with controlling variation within location or within time period. This can be handled by using resolvable block designs. Here, locations or time periods may be taken as replications and the variation within a location or a time period can be taken care of by blocking within the replication.

In an agricultural field experiment, the land may be divided into a number of large areas corresponding to the replications and then each area may be subdivided into incomplete blocks. These designs are also quite useful for varietal trials conducted in the NARS and help in improving the precision of treatment comparisons.

Square Lattice Design and the Rectangular Lattice Design are resolvable PBIB designs. These designs are very useful to the experimenters for conducting their experiments and can be used both in initial and in advanced varietal trials. These designs, however, demand that the number of treatments v satisfy the restriction $v = s^2$ or $v = s(s-1)$, where s is a prime number. A more general class of resolvable incomplete block designs is α -designs. These designs are available for $v = pq$, p and q being positive integers. Thus, α -designs are available whenever the number of treatments is a composite number. Square Lattice and Rectangular Lattice designs are special cases of α -designs.

A big stumbling block to the experimenters in the adoption of these designs is the fact that the layout of these designs is not readily available. The experimenters also feel that the analysis of the data generated from these designs and interpretation of results may be a problem. However, with the advent of statistical software packages, the analysis of data generated from any designed experiment is no longer a problem.

There is, therefore, a need to prepare a catalogue of alpha designs giving the layout of the design so that the experimenters can make use of these important designs in their experiments. These designs are incomplete block designs and their generation is not unique. It is, therefore, important to generate an alpha design that has some optimality property or has high efficiency. The authors have addressed this problem also and have reported designs with high A-efficiency.

The authors do not claim that the designs presented are the most efficient. Indeed it may be possible to generate an α -design with more A-efficiency than a corresponding design included in this monograph. However, the designs reported have relatively high A-efficiency.

The present monograph on α -designs is an attempt by the authors to provide a good catalog of nice α -designs. The authors have provided a ready-made solution to the experimenters for planning and designing their experiments. The designs included in the monograph are, however, restricted to the case where the number of treatments is smaller than 150, the number of replications is smaller than 5, and the block size is smaller than 10. As mentioned before, the designs have high A-efficiency.

The authors have to be congratulated for doing a nice job which is sure to be very helpful to experimenters and practicing statisticians.

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FOREWORD

The Indian Agricultural Statistics Research Institute is a premier Institute in the discipline of Agricultural Statistics and Computer Applications in the country. The Institute has been engaged in conducting research and organizing teaching and training programmes in Agricultural Statistics with special emphasis on Experimental Designs, Sampling Techniques, Statistical Genetics, Crop Forecasting Techniques, Econometrics, Statistical Computing and Computer Applications. For disseminating the research in Agricultural Statistics and Computer Applications to the scientists of the National Agricultural Research System for helping them improve the quality of agricultural research, the Institute has taken a lead in developing Statistical Software Packages for generating designs and analysis of data.

Designing an experiment is an essential component of every scientific investigation in agriculture and allied sciences. It is through the data collected from designed experiments that valid inferences are drawn. The institute has made many significant contributions in the field of experimental designs. In order to make research globally competitive, it is essential that sound statistical methodologies be adopted for the collection and analysis of data.

The Institute has established itself in the world of Statistics so far as research in Experimental Designs is concerned. Many notable contributions have been made in the field of experimental designs at this Institute and a number of research papers have appeared in international statistical journals of repute. As many as 4 textbooks and one handbook on Design of Experiments have been published by the scientists of this Institute. An electronic book on Design and Analysis of Agricultural Experiments has also been developed. The scientists of the Institute have also developed a website 'DESIGN RESOURCES' for the benefit of scientists all over the world (<http://www.iasri.res.in/design>). The main objective of this design resources server is to develop a WEB DESIGN in NARS.

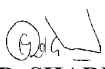
The Institute has also become a veteran in rigorously undertaking the consultancy and advisory services for the agricultural research personnel. The Institute has a vast experience of organizing several training programmes for the scientists of National Agricultural research System (NARS) with a view to improve the experimentation by adopting efficient, cost effective and robust designs and to use sound, sophisticated analytical techniques for the statistical analysis of data.

Encouraged with the adoption of efficient designs and sophisticated analytical techniques, we have thought of bring out monographs on topics of current interest. To make research in agricultural sciences more meaningful using sophisticated statistical methodologies for analyses, user friendly monographs are very useful.

The present monograph on α -designs is a valiant attempt in providing the layout of the resolvable designs to the experimenters and practicing statistician. The α -designs are resolvable block designs and are quite useful for controlling the heterogeneity in the experimental material and simultaneously retaining the demonstration property of the treatment response in agricultural field experiments especially in varietal trials. Seeing the high percentage of the experiments with smaller block mean squares as compared to error mean square and high CV in the National Agricultural Research System. This monograph on α -designs will be quite useful for improving the status of experimentation in National Agricultural Research System.

This monograph will serve as a ready reckoner to the experimenters and practicing statisticians. I take this opportunity to congratulate the authors for doing a wonderful job.

New Delhi
February 17, 2007


(S.D. SHARMA)
Director, IASRI

PREFACE

This monograph on α -designs is an outcome of the various discussions held and advisory services provided to scientists in the National Agricultural Research System. During a workshop of the All India Co-ordinated Research Project on Rapeseed and Mustard; National Research Centre on Rapeseed and Mustard, Bharatpur, held at Pant Nagar, through the presentations and the discussions held, it was realized that α -designs would be very useful for the experiments conducted on Rapeseed and Mustard varietal trials. In fact these designs will be very useful for any varietal trial. α -designs are essentially resolvable designs with each resolvable group a complete replicate. But each replicate comprises of several blocks and randomization of treatments is made within each block in the resolvable group. On the face of it, these designs look like Randomized Complete Block (RCB) designs, but these are not matter-of-factly RCB designs. In RCB design, the randomization of treatments is done within each of the blocks in the replication. A large number of experiments are conducted in this fashion, but the randomization of treatments is not done as per the requirement of the α -designs. However, if the experimenters modify the randomization process of treatments, they can derive a lot of useful information from such experiments and improve the precision of comparisons reducing thereby considerably the coefficient of variation (CV) of the experiment. Discussions on the usefulness of these designs were held during the meeting of the Committee constituted to finalize evaluation of Bt. Cotton entries into AICRP's. As a follow up action scientists from Division of Crop Improvement, Central Institute of Cotton Research, Nagpur, were advised on the designing of four experiments to be conducted with cotton varieties. First two of these experiments are to be conducted to morphologically characterize varieties and hybrids with respect to about 45 morphological characters both qualitative and quantitative and document the database as a part of Distinctness, Uniformity and Stability (DUS) testing.

However, it may not be possible for the experimenters to lay their hands on these α -designs. The construction of these designs is not easy. An experimenter has to get associated with a statistician to get a randomized layout of this design. For the benefit of the experimenters, we got motivated to prepare a catalogue of α -designs for reasonable number of treatments and replications. In this catalogue we have restricted to number of treatments less than or equal to 150, number of replications less than or equal to 5 and the block size less than or equal to 10. For the benefit of experimenters, complete layout of the designs is given. These designs have been generated by using an intelligent computer algorithm. The designs reported are highly efficient. Indeed it is possible to generate a design for given parameters with a better efficiency than that of the design reported. Many designs generated are already known in the literature.

The authors wish to thank all the scientists of NARS with whom we had several useful discussions. It is expected that this catalogue will prove to be of immense value to the experimenters in their experimentation. The authors are also indebted to all the scientists, students and research associates who have helped directly or indirectly in the preparation of this monograph. The authors also wish to thank Dr. SD Sharma, Director, IASRI for providing facilities and encouraging us to prepare this monograph.

Any suggestions for improvement of the presentation of the material are welcome.

-Authors

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α -Designs

1. Introduction

Block designs are useful in one-way elimination of heterogeneity settings. A block design is said to be complete, if it contains every treatment once in each block. A block design is said to be incomplete if it has at least one block that does not contain all the treatments. Most commonly used block designs are randomized complete block (RCB) designs, balanced incomplete block (BIB) designs, partially balanced incomplete block (PBIB) designs, lattice designs, etc. In block designs, the idea is to form blocks in a manner such that experimental units within a block are as homogeneous as possible, so that between block variations can be isolated from the error component. This results into reduction of the error component, which in turn enhances the precision of the comparison of treatment effects.

Another useful concept in the context of incomplete block designs is that of resolvability wherein, the blocks can be grouped such that with in each group, every treatment appears exactly once. This concept is due to Bose (1942) who defined it for BIB designs. Resolvable block designs have a lot of practical significance as these designs allow performing an experiment one replication at a time. For example, field trials with large number of crop varieties cannot always be laid out in a single location or a single season. Therefore, it is desired that variation due to location or time periods may also be controlled alongwith controlling within location or time period variation. This can be handled by using resolvable block designs. Here, locations or time periods may be taken as replications and the variation within a location or a time period can be taken care of by blocking. In some experimental situations, it may not be feasible to run the entire experiment in one session. The experiment may have to be run in different sessions. Resolvable designs become useful in this situation. A single replication may be run in one session. If an experiment has to be discontinued at any point of time then all the treatments would have occurred equally often (provided the design is connected upto that session). Accuracy will also be increased if the experimental material can be arranged so that replicates are relatively homogeneous. In an agricultural experiment, for example, the land may be divided into a number of large areas corresponding to the replications and then each area is subdivided into blocks.

In National Agricultural Research System (NARS), a large number of experiments are conducted for improvement of crop varieties. A large number of varieties in these trials are fixed, *i.e.*, not at the choice of statistician, and are large enough to require use of incomplete block designs or some other method of controlling the variability in the experimental units. Number of replications are also fixed based on the availability of land and other experimental resources. Further, it has been seen that in many crop improvement programmes, initial varietal trials and advanced varietal trials are generally conducted using a RCB design. The analysis of the data reveals that the coefficient of variation (CV) in many of these experiments is high and as a consequence, the precision of treatment comparisons is low. This amounts to saying that the error sum of squares is large compared to the sum of squares attributable to the model and hence, small treatment differences may not be detected as significant. A large number of trials are rejected due to high CV.

It is hypothesized that the fundamental problem in these experiments is that the between block variations are non-significant in a large number of the cases. In any experimentation, non-significance of block effects or a high value of CV may arise due to:

1. Bad management of the controllable factors during experimentation (managerial aspects)
2. Faulty formation of blocks (designing)
3. Lack of identification of ancillary information that could have been used as covariate (analysis of covariance).

The first point may be taken care of if the experimenter is very cautious and experienced. Analysis of covariance is an analytical procedure and is very effective in controlling the experimental error although it has nothing to do with the designing of the experiment. The most important point which has to be taken care of during the allocation of treatments to different experimental units is to adopt proper blocking techniques. In field trials, the formation of blocks, *i.e.*, shape and size should be preceded by uniformity trials to determine the fertility pattern through a contour map. In most situations particularly in the plains, the contour maps show the variations in the fertility patches rather than in a single direction. Therefore, there is a need to effectively control the variation through blocking.

This necessitates the use of incomplete block designs like BIB designs, PBIB designs, cyclic designs, etc. One may, however, argue that in these designs the purpose of demonstration of a variety effect in the field cannot be done as all the treatments are not appearing in adjacent piece of land. To overcome this problem, it is recommended that resolvable block designs with smaller block sizes may be used. We now give some definitions and examples relating to resolvable block designs.

Definition 1.1: [Bose, (1942)]. A BIB design is called **resolvable** if the b blocks of a BIB design can be divided into r groups such that each of the v treatments occur exactly once in each group. Clearly, $b/r = x$, where x is the number of blocks in each resolvable group.

The concept of resolvability was extended to other block designs like PBIB designs and cyclic designs. In general, a proper block design is said to be resolvable if its b blocks can be grouped into r sets of blocks, each set containing $b/r = x$ blocks, such that every treatment appears in each set precisely once.

Example 1.1: Following BIB design with parameters $v = 4$, $b = 6$, $r = 3$, $k = 2$, $\lambda = 1$ is resolvable. The columns represent blocks and the symbols within blocks are treatments in the block.

Replication I		Replication II		Replication III	
Block-I	Block-II	Block-I	Block-II	Block-I	Block-II
1	3	1	2	1	2
2	4	3	4	4	3

Example 1.2: Following group divisible design with parameters $v = b = 4$, $r = k = 2$, $\lambda_1 = 0$, $\lambda_2 = 1$, $m = 2$, $n = 2$ is resolvable. The columns represent blocks and the symbols within blocks are treatments in the block.

Replication-I		Replication-II	
Block-I	Block-II	Block-I	Block-II
1	3	1	2
2	4	3	4

Example 1.3: An experimenter is interested in comparing 12 genotypes. A total of 36 experimental units are available that can be arranged in 6 blocks of 6 plots each. A resolvable block design with block contents is:

Replication-I		Replication-II		Replication -III	
Block-I	Block-II	Block-I	Block-II	Block-I	Block-II
1	3	1	2	1	2
2	4	3	4	4	3
7	5	6	5	6	5
8	6	8	7	7	8
9	10	11	9	9	11
11	12	12	10	10	12

Resolvable block designs have been studied by various authors in the literature. Lattice designs introduced by Yates (1936) especially for varietal trials are resolvable. Lattice designs are available for a limited number of varieties and block sizes. The simple ($r = 2$) and triple lattices ($r = 3$) require v , the number of varieties, to be of the order s^2 , the number of blocks in each replication and the block size k is s . Further conditions are imposed on v in quadruple and higher order lattices. Harshbarger (1949) extended the principle of square lattice designs to simple and triple rectangular lattice designs with $v = s(s-1)$ and $k = s-1$.

Bose and Nair (1962) have described and tabulated a series of two-replicate resolvable designs. These designs usefully augment the simple, square and rectangular lattices. David (1967) observed that cyclic designs are also resolvable designs but under the restrictions that k is multiple of r . For an excellent literature on cyclic and computer generated designs a reference may be made to John and Williams (1995) and references cited therein.

Kageyama (1972) gave a list of resolvable BIB designs. Clatworthy (1973) provided information on the resolvability of the two associate PBIB designs. Parsad et al. (2000) provided a comprehensive list of resolvable binary balanced block designs including BIB designs. Nigam et al. (2006) gave the catalogues of resolvable BIB designs and binary balanced block designs for $v \leq 10$ alongwith their block contents. They have also shown that resolvable block designs are very useful for on-farm trials.

In agricultural experiments, particularly concerning varietal trials, a large number of varieties are to be tested simultaneously, thus increasing the size of experimentation. For efficient experimentation it is desirable to conduct these trials using resolvable block designs. As number of varieties is not fixed and experimenter has to choose number of replications as per experimental resources, several efficient designs such as Lattice designs, BIB designs, cyclic designs etc. may not meet the requirements of the experimenter.

To meet up the requirements of the experimenter, it is desirable to generate special kind of resolvable designs, called α -designs, given by Patterson and Williams (1976). It is a wide class of resolvable block designs for which v and b have common integer multiple $s \geq 2$. Clearly $v = ks$ and $b = rs$, k , r and s are the parameters of α -design. In the sequel we describe the construction of α -designs.

2. Construction of α -designs

For constructing α -designs for $v = ks$ treatments, in r replications we start with $k \times r$ array called α -array with element integer modulo s . An α -design is completely determined by an $k \times r$ α -array. By each column of this array we develop one replication of s -blocks each of size k . Therefore, we have r -replications of rs -blocks of size k each. For the sake of completeness, we define the following:

Array: Arrangement of elements in rows and columns is called array. An array taken for developing designs is called generating array.

Reduced array: Generating array with all elements equal to zero in the first row and first column is called as the reduced array.

Under a suitable relabelling or permutation of elements, all arrays can be represented as reduced arrays, thereby restricting the search of designs to reduced arrays only. For example, consider the following generating array for $s = 3$

1	0	1
2	2	1
0	1	1
1	0	2

First add 2 to the elements of columns 1 and 3 reducing the sum to modulo 3 when necessary. We get the following array.

0	0	0
1	2	0
2	1	0
0	0	1

Now add 2 to the elements in row 2 and 1 to those in row 3 and reduce each element modulo 3 when necessary. The resulting array is the reduced array used for this example. The reduced array can be generated by arbitrary allocation of symbols in a $k \times r$ array modulo s by fixing the entries in the first row and first column as zero. As far as possible all the $r-1$ columns of $k-1$ size each should be distinct. The design obtainable from this array may then be checked for A-efficiency. If the A-efficiency is poor, then one can select another reduced array, till the design with desired A-efficiency is obtained. Indian Agricultural Statistics Research Institute (IASRI) has developed a computer algorithm based on exchange and interchange procedure to generate efficient α -designs. The designs presented here are obtained through the computer algorithm, trial and error solutions or from the arrays available in the literature. The procedure of generation of α -designs from reduced α -array and terminology involved are described in the sequel.

1. **α -array:** It is an $k \times r$ array of elements consisting of residue class of s , i.e., the elements can be $0, 1, 2, \dots, s-1$. If the first row and first column of that array consists of zeros as elements, then that array is called reduced array. We will describe the construction of the α -designs by the help of the reduced array. We can determine the concurrences of the different treatment pairs by the α -array.
2. **Intermediate Array:** Construction of that intermediate array is the intermediate step to develop α -design. One can have $(s-1)$ new columns from a column of α -array by developing the column cyclically modulo s . By this way we can construct $r(s-1)$ new columns by cyclically developing each column of the α -array modulo s . This arrangement of $k \times sr$ array is called intermediate array.
3. **α -Design:** Now, take the intermediate array and add $(1 + (j-1)s)$ to the elements of j^{th} row of that array ($j = 1, 2, \dots, k$). This gives us an array of size $(k \times sr)$. Now taking one to one correspondence between the treatments and the elements of the final array, we get an α -design with the parameters $v = ks, b = rs, r, k$.

The steps of construction are described with the help of following example:

Example 2.1: Consider the construction of an α -design for $v = 12, b = 12, r = 3, k = 3, s = 4$. The generating array α for this design is

0	0	0
0	2	3
0	3	1

Given this generating array, develop the intermediate array α^* by cyclically developing each column modulo s . The intermediate array α^* is

0	1	2	3	0	1	2	3	0	1	2	3
0	1	2	3	2	3	0	1	3	0	1	2
0	1	2	3	3	0	1	2	1	2	3	0

Now add $(1 + (j-1)s)$ to the elements of j^{th} row of that array ($j = 1, 2, \dots, k$). The α -design with columns as blocks is

Replication-I				Replication-II				Replication-III			
B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12
1	2	3	4	1	2	3	4	1	2	3	4
5	6	7	8	7	8	5	6	8	5	6	7
9	10	11	12	12	9	10	11	10	11	12	9

B#: denotes the block number

We are developing α -design from the α -array by cyclically developing modulo s the initial blocks corresponding to a column of α -array and adding $1 + (j-1)s$ to the j^{th} place of each of

the developed blocks. Therefore, some of the concurrences, at least $k \binom{s}{2}$ are zero. To be clearer, consider the following example:

Example 2.2: Consider an 3×4 α -array with $k = 3, r = 4, s = 3$ as

$$\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 \\ 0 & 2 & 1 & 2 \end{array}$$

Take 3rd column as initial set: then developing cyclically modulo 3 we get

$$\begin{array}{ccc} 0 & 1 & 2 \\ 2 & 0 & 1 \\ 1 & 2 & 0 \end{array}$$

Now add $1 + (j - 1)s = 1 + 3(j - 1)$ to the elements in the j^{th} row for $j = 1, 2, 3$:

$$\begin{array}{ccc} 1 & 2 & 3 \\ 6 & 4 & 5 \\ 8 & 9 & 7 \end{array}$$

The treatments 1, 2, 3; 4, 5, 6 and 7, 8, 9 will appear in different blocks in any of the replications and hence never occur together in a block.

In some cases, the replications get repeated in α -designs. The conditions for repeated replications in α -designs are described in the sequel. A typical α -design is generated by using the reduced array or α -array. In that array, the first row and first column is a zero-vector. The α -array constitutes the residue class of s as elements. Therefore, we have to choose $(k - 1)$ elements from s -elements to form a column of the α -array. Distinct choice of these columns gives different replications.

Let number of distinct columns of size $(k - 1)$ produced from the modulo residue s be $m = s^{(k-1)}$. Out of these total number of distinct columns, one column will be of all zeroes that is already taken as the first column of the α -array. Therefore, we have to select $(r - 1)$ distinct columns out of remaining $(m - 1)$ columns. Therefore, if $r - 1 > m - 1$ or $r > m$, repetition of a column in the α -array is must.

To be clearer, consider the following example:

Example 2.3: Consider a α -design with parameters $v = 6, b = 10, r = 5, k = 3$ to be obtained from 3×5 α -array having elements of modulo class 2, i.e., $s = 2$. The elements of modulo class 2 are 0 and 1. Each replication of the array is obtained from columns of the α -array. In a column of an α -array the first element is always 0. Therefore, from the remaining $(k - 1) = 2$ elements

we have to choose out of $s = 2$ elements, i.e., $\{0, 1\}$. Different choices of such is $s^{k-1} = 2^2 = 4$, i.e., $(0,0)$; $(0,1)$; $(1,0)$; $(1,1)$. Now we can get 4-distinct replications by choosing the 4-columns of the α -arrays as

$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
0	0	0	0
0	0	1	1
0	1	0	1

Each of the above column is used to get different replications of the α -design. Now 5th column of the α -array can be either one of them. Therefore, one replication always gets repeated to construct 5-replicate α -design for $k=3$ and $s = 2$.

Dual of an α -design

It is interesting to note that the dual of an α -design is also an α -design. Williams (1975) has shown that the dual of an α -design obtained through an $k \times r$, α -array can be obtained by generating an $r \times k$ array from $k \times r$ α -array with elements as $s - \alpha(q, p)$ modulo s , where $\alpha(p, q)$ are the elements of the generating array α . To make the exposition clear, consider the following example.

Example 2.4: Consider the 3×4 α -array with $k = 3$, $r = 4$, $s = 3$ given in Example 2.2 as

0	0	0	0
0	1	2	2
0	2	1	2

Now dual of this array is a 4×3 array with $k^* = 4$, $r^* = 3$, $s^* = 3$ with elements as

0	0	0
0	2	1
0	1	2
0	1	1

3. Randomization of α -design

For an α -design, there are four stages of randomization viz.

- i) Randomize the treatments, i.e., Random allocation of treatments to treatment numbers.
- ii) Randomize the replications
- iii) Randomize the blocks within each replication. Separate randomization should be done for each replication.
- iv) Randomize the treatments to experimental units within each block. Separate randomization should be done for each of the blocks.

The randomization of the α -design is done by generating random numbers or by the random number tables. This is explained through the randomization of α -design given in Example 2.1.

Step 1: Assume that the treatments have been numbered randomly.

Step 2: Randomize the replicates.

Generate random numbers 1 to 3. Let the numbers be {2, 1 and 3}. Then arrange the replication 2 as replication 1; replication 1 as replication 2 and replication 3 as replication 3. The layout of the design after implementation of this step is:

Replication-I				Replication-II				Replication-III			
B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12
1	2	3	4	1	2	3	4	1	2	3	4
7	8	9	6	5	6	7	8	8	5	6	7
12	9	10	11	9	10	11	12	10	11	12	9

Step 3: Randomize the blocks.

Generate 3 sets of 4-distinct random numbers less than equal to 4. Let the numbers be {3, 1, 2, 4}, {3, 2, 4, 1} and {4, 1, 2, 3}. Randomize each of the blocks by these sets of the random numbers. Use first set to randomize the blocks within replication 1, second set to randomize the blocks of replication 2 and use third set to randomize the blocks within replication 3. The resulting layout of the design is:

Replication-I				Replication-II				Replication-III			
B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12
3	1	2	4	3	2	4	1	4	1	2	3
9	7	8	6	7	6	8	5	7	8	5	6
10	12	9	11	11	10	12	9	9	10	11	12

Step 4: Randomize the treatments within each block.

Generate 12 sets of 3-random numbers each. Let the sets be {1, 3, 2}; {3, 1, 2}; {3, 2, 1}; {2, 1, 3}; {1, 2, 3}; {3, 1, 2}; {2, 3, 1}; {3, 2, 1}; {3, 2, 1}; {2, 1, 3}; {1, 3, 2}; {3, 1, 2}. Use these sets to randomize the treatments within blocks.

Replication-I				Replication-II				Replication-III			
B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12
3	12	9	6	3	10	8	9	9	8	2	12
10	1	8	4	7	2	12	5	7	1	11	3
9	7	2	11	11	6	4	1	4	10	5	6

At the end of the step 4 we get randomized layout plan of the α -design. This plan can now be implemented for experimentation.

4. Lower bounds to A- and D-efficiency of the designs

Different α -designs with given parameters v , k , r can be obtained from different generating arrays. The designs would have different A-and D-efficiencies. The choice of an appropriate design can thus be based on the lower bound to the average efficiency of the designs.

Let $\mathbf{D}(v, b, k)$ denotes the class of connected block designs in which v treatments are arranged in b blocks each of size k . A design $d^* \in \mathbf{D}(v, b, k)$ is said to be A-optimal if it has minimum value of trace of variance-covariance matrix of the best linear unbiased estimate (BLUE) of treatment contrasts of interest over all possible combinatorial solutions of designs $\mathbf{D}(v, b, k)$. Similarly a design $d^* \in \mathbf{D}(v, b, k)$ is said to be D-optimal if it minimizes the determinant of the variance-covariance matrix of BLUE of treatment contrasts of interest.

Resolvable block designs have a nested blocking structure, i.e., the incomplete blocks within each replicate are randomized separately for each replicate. Such randomization process is appropriate for an experimental layout where the replicates are separate entities. Thus, the blocks of one replicate have no relation to those of another. However, for computation of efficiency, we can make use of the results of general block designs. We shall now discuss the concept of lower bound of the A-[D-] efficiency due to Rathore (2004) and Rathore *et al.* (2006). They obtained lower bound to A-[D-] efficiency of $d \in \mathbf{D}(v, b, k)$; the class of connected block designs in which v treatments are arranged in b blocks each of size k . These bounds are obtained on the lines of Cheng and Wu (1981).

It can easily be shown that the problem of obtaining an A-[D-] optimal design for making all possible pairwise treatment comparisons is equivalent to the problem of obtaining an A-[D-] optimal design for a complete set of orthonormal treatment contrasts $\mathbf{P}'\boldsymbol{\tau}$; $\mathbf{P}'\mathbf{P} = \mathbf{I}_{v-1}$, $\mathbf{P}\mathbf{P}' = \mathbf{I}_v - \mathbf{1}\mathbf{1}'/v$. For inferring on complete set of orthonormal treatment contrasts, a design is said to be A-optimal if it minimizes the sum of reciprocals of the non-zero eigenvalues of the $\mathbf{C}$ -matrix over $\mathbf{D}(v, b, k)$ and is said to be D-optimal if it minimizes the inverse of the product of the nonzero eigenvalues of $\mathbf{C}$ -matrix over $\mathbf{D}(v, b, k)$. For a connected block design d ,

let $\theta_1, \theta_2, \dots, \theta_{v-1}$ be the non-zero eigenvalues of $\mathbf{C}$. Now define $\phi_A(d) = \sum_{i=1}^{v-1} \theta_i^{-1}$ and

$\phi_D(d) = \prod_{i=1}^{v-1} \theta_i^{-1}$. Then, a design is A- [D-] optimal if it minimizes the $\phi_A(d)[\phi_D(d)]$ over $\mathbf{D}(v, b, k)$. The A-efficiency $\{e_A(d)\}$ and D-efficiency $\{e_D(d)\}$ of any design d over $\mathbf{D}(v, b, k)$ is defined as

$$e_A(d) = \frac{\phi_A(d_A^*)}{\phi_A(d)} \text{ and } e_D(d) = \left[\frac{\phi_D(d_D^*)}{\phi_D(d)} \right]^{1/(v-1)} \quad (3.1)$$

where, d_A^* and d_D^* are the A-optimal and D-optimal design over $\mathbf{D}(v, b, k)$, respectively.

Using the arithmetic, harmonic and geometric means inequality, and following the lines of Cheng and Wu (1981), the A-efficiency $\{e_A(d)\}$ and D-efficiency $\{e_D(d)\}$ of any design d over $\mathbf{D}(v, b, k)$ are given by

$$e'_A(d) = \frac{(v-1)^2}{\{b(k-1)\phi_A(d)\}} \quad \text{and} \quad (3.2)$$

$$e'_D(d) = \frac{(v-1)}{[b(k-1)\{\phi_D(d)\}^{1/(v-1)}]} \quad (3.3)$$

It may be noted here that Cheng and Wu (1981) obtained lower bounds to the A- and D-efficiencies in the class of connected binary proper block designs but here the lower bounds correspond to the class of connected proper block designs. This is because for any proper block design d , $\text{trace}(\mathbf{C}_d) \leq b(k-1)$.

If the given design d is a BIB design, then, $\phi_A(d) = k(v-1)/\lambda v$ and $\phi_D(d) = (k/\lambda v)^{(v-1)}$. For a BIB design, $\lambda = r(k-1)/(v-1)$ and $vr = bk$. Therefore, $\phi_A(d) = k(v-1)^2/\{vr(k-1)\} = (v-1)^2/\{b(k-1)\}$ and $\phi_D(d) = \{k(v-1)/vr(k-1)\}^{(v-1)} = ((v-1)/b(k-1))^{(v-1)}$. Clearly for a BIB design both e'_A and e'_D are equal to 1.0000.

5. Catalogue of α -designs

Here, the α -designs with high values of lower bounds to A-efficiency and D-efficiency in the class $\mathbf{D}(v, b, k)$ are obtained and presented. We have prepared a comprehensive list of α -designs for $v \leq 150, k \leq 10, r \leq 5$ and $s \leq 15$ along with lower bounds to A- and D-efficiencies; concurrences and generating arrays. The layout of these designs along with block contents has also been prepared. The distribution of designs in different efficiency ranges is:

Frequency Class	<0.8000	0.8000-0.8500	0.8500-0.9000	0.9000-0.9500	0.9500-0.9750	0.9750-0.9900	>0.9900	Total
A-Efficiency	18	17	43	116	151	102	1	448
D-Efficiency	0	5	10	49	123	203	58	448

The catalogue of parameters of α -designs along with lower bounds to A- and D-efficiencies and concurrences is given in Table 1 (Appendix). The generating α -arrays are given in Table 2 (Appendix). The block contents of the designs are given in Table 3 (Appendix). In Tables 1 and 3, the resolvable partially balanced incomplete block designs with 2-associate classes {PBIB(2)} designs of Clatworthy (1973); Square Lattice designs and Rectangular Lattice designs that have equal lower bounds to A- and D-efficiencies to that of corresponding α -designs are also given in remarks column.

5.1 Comparison of the new designs with existing designs

The designs obtained have been compared with corresponding square lattice designs, rectangular lattice designs, resolvable PBIB(2) designs given in Clatworthy (1973) and the α -designs obtainable from basic arrays given by Patterson et al. (1978) and from dualization of these basic arrays. The more efficient designs have been marked with *** in Tables 1 and 3.

Eleven designs are more efficient than the corresponding resolvable PBIB (2) designs (S11, S38, S69, S114, LS8, LS30, LS54, LS76, LS89, LS126 and LS140). These PBIB(2) designs are given at serial numbers 1 to 11 in Table 4 (Appendix). It is interesting to note here that for the PBIB(2) designs based on Latin square association scheme the concurrences of the treatments were 0 or 2 and for singular group divisible designs the concurrences were either 1 or 5. Further all the designs LS8, LS30, LS54, LS76, LS89, LS126 and LS140 were obtained by taking two copies of a design with 2-replications.

Patterson et al. (1978) gave 11 basic arrays which provide 33 α -designs for $v = 25, k = 5, r = 2, 3, 4$; $v = 36, k = 6, r = 2, 3, 4$; $v = 49, k = 7, r = 2, 3, 4$; $v = 64, k = 8, r = 2, 3, 4$; $v = 81, k = 9, r = 2, 3, 4$; $v = 99, k = 9, r = 2, 3, 4$; $v = 96, k = 8, r = 2, 3, 4$; $v = 91, k = 7, r = 2, 3, 4$; $v = 98, k = 7, r = 2, 3, 4$ and $v = 90, k = 6, r = 2, 3, 4$.

The designs with $r = 4$ can be obtained from the basic array itself. The designs with $r = 3$ ($r = 2$) can be obtained by taking any of the 3 columns (2 columns) of the basic arrays. We obtained designs with all possible combinations and listed designs with highest lower bound to A-efficiency for each of the parametric combinations. These listed designs were then compared with the designs generated in the present investigation. 10 designs were found to be more efficient than the designs obtainable from basic arrays. Less efficient α -designs designs are given from serial numbers 12 to 21 in Table 4.

Further 88 more α -designs with $k = 3$ and $k = 4$ can be obtained from dual of these 11 basic arrays. To generate α -design for $k = 4$ from the dual of the basic array of Patterson et al. (1978), we have taken first 5 rows to generate 5-replicate designs, first 4-rows to generate 4-replicate designs, first 3-rows to generate 3-replicate designs and first 2-rows of the dual of the array to get 2-replicate designs.

To get efficient α -designs for $k = 3$, we have taken most efficient combination of 3-columns out of 4-columns of the subset of the basic array that is used to generate α -design for $k = 4$. 48 designs (29 with $k = 4$ and 19 with $k = 3$) are more efficient than the designs obtainable by dualization of basic arrays. These designs are given at serial numbers from 22 to 69 in Table 4.

25 designs have been obtained for which no corresponding resolvable solution of PBIB(2) designs is available in literature. The list of corresponding PBIB(2) designs is S28, S86, SR18, SR41, SR52, SR58, SR66, SR75, SR80, R42, R70, R97, R109, R139, T14, T16, T20, T44, T48, T49, T72, T73, T86, T87 and M16. The parameters of the non-resolvable PBIB (2) designs are given in Table 5 (Appendix). The α -designs corresponding to these designs are also marked as \$ in Tables 1 and 3.

There are 32 resolvable designs available in the literature (18 PBIB(2) designs, 10 square lattice designs and 4 rectangular lattice designs) that are more efficient than the corresponding α -designs. The parameters of the more efficient resolvable block designs are given in Table 6 (Appendix).

Throughout Tables 1, 3, 4, 5 and 6, the designs X# denotes the design of Type X at serial number # in Clatworthy (1973).

6. Discussion

In some cases, the treatments may be partitioned into two sets. The one set consists of the new treatments/varieties/entries or new material called test treatments and other set may consists of the already existing treatments or check varieties, called control treatments. The experimenter is interested in verifying whether or not any of the test treatments can replace the control treatments. Initial varietal trials (IVT) and advanced varietal trial-I (AVT-I) are common examples of such trials. In IVT the material/entry from the breeders is taken and is compared with already released varieties or check varieties. Material that is found superior to the check varieties is promoted to AVT-I. Again in AVT-I, the entries promoted to AVT-I are compared with best performing check. For such experimental situations, one may take an α -design in test treatments or new entries and reinforce each block of the design by the control treatments or check varieties. Such designs are known as reinforced α -designs.

For multi-location trials, if there are some entries which are common to all environments and there are some additional entries then one may select an α -design for the number of entries that are common to all the environments and reinforcing the additional environment specific additional entries to all the blocks within an environment.

7. Analysis

The analysis of α -designs can be performed as per procedure of general block designs. The only change with respect to analysis is that the block sum of squares will be bifurcated into sum of squares due to replications and sum of squares between blocks within replications. As replications versus blocks (within replications) and replications versus treatments classifications

are orthogonal, therefore, sum of squares due to replications is given by $\sum_{j=1}^r \frac{R_j^2}{v} - CF$, where r is

the number of replications. Blocks within replication sum of squares can be obtained by subtraction and both these classifications are tested against mean square error. An outline of the analysis of variance Table is given as

ANOVA

Source of Variation	Degrees of Freedom
Replications	$r - 1$
Blocks within replications	$r(s-1)$
Treatments	$v-1$ or $sk-1$
Error	$rsk-sk-rs+1$
Total	$vr-1$ or $rsk-1$

All possible pairwise treatment (entries) comparisons can be made easily. The entries that are significantly different from the check and have higher adjusted means are promoted to the next stage.

Summary

α -designs are essentially resolvable block designs. In a resolvable block design, the blocks can be grouped such that in each group, every treatment appears exactly once. Resolvable block designs allow performing an experiment one replication at a time. For example, field trials with large number of crop varieties cannot always be laid out in a single location or a single season. Therefore, it is desired that variation due to location or time periods may also be controlled along with controlling within location or time period variation. This can be handled by using resolvable block designs. Here, locations or time periods may be taken as replications and the variation within a location or a time period can be taken care of by blocking. In an agricultural experiment, for example, the land may be divided into a number of large areas corresponding to the replications and then each area is subdivided into blocks. These designs are also quite useful for varietal trials conducted in the NARS and will help in improving the precision of treatment comparisons. A critical look at the experimentation in the NARS reveals that alpha designs have not found much favour from the experimenters. It may possibly be due to the fact that the experimenters may find it difficult to lay their hands on these α -designs. The construction of these designs is not easy. An experimenter has to get associated with a statistician to get a randomized layout of this design. For the benefit of the experimenters, a comprehensive catalogue of α -designs for $6 \leq v(=sk) \leq 150$, $2 \leq r \leq 5$, $3 \leq k \leq 10$ and $2 \leq s \leq 15$ has been prepared along with lower bounds to A- and D- efficiencies and generating arrays.

The layout of these designs along with block contents has also been prepared. The designs obtained have been compared with corresponding Square Lattice, Rectangular Lattice, Resolvable two-associate Partially Balanced Incomplete Block {PBIB (2)} designs and the α -designs obtainable from basic arrays given by Patterson et al. (1978). Eleven designs are more efficient than the corresponding resolvable PBIB (2) designs (S11, S38, S69, S114, LS8, LS30, LS54, LS76, LS89, LS126 and LS140). It is interesting to note here that for the PBIB (2) designs based on Latin square association scheme, the concurrences of the treatments were 0 or 2 and for singular group divisible designs the concurrences are either 1 or 5. Further all the designs LS8, LS30, LS54, LS76, LS89, LS126 and LS140 were obtained by taking two copies of a design with 2-replications. 10 designs were found to be more efficient than the designs obtainable from basic arrays. 48 designs (29 with $k = 4$ and 19 with $k = 3$) are more efficient than the designs obtainable by dualization of basic arrays. 25 designs have been obtained for which no corresponding resolvable solution of PBIB(2) designs is available in literature. The list of corresponding resolvable PBIB(2) designs is S28, S86, SR18, SR41, SR52, SR58, SR66, SR75, SR80, R42, R70, R97, R109, R139, T14, T16, T20, T44, T48, T49, T72, T73, T86, T87 and M16. Here X# denotes the design of type X at serial number # in Clatworthy(1973).

Lkkjk'k

यह मोनोग्राफ एक विशेष प्रकार की रिजॉल्वेबल अपूर्ण ब्लॉक अभिकल्पनाओं से संबंधित है जिन्हें एल्फा अभिकल्पनाएँ कहते हैं। रिजॉल्वेबल ब्लॉक अभिकल्पना एक ऐसी ब्लॉक अभिकल्पना होती है जिसमें ब्लॉकों को इस तरह से वर्गीकृत किया जा सकता है कि ब्लॉकों के प्रत्येक समूह में प्रत्येक ट्रीटमेंट केवल एक **ckj vkrk gSA** रिजॉल्वेबल ब्लॉक अभिकल्पनाओं की सहायता से एक पुनरावृत्ति को एक बार लेकर भी परीक्षण किये जा सकते हैं। उदाहरणतया, कृषि खेत परीक्षणों में फसलों की बड़ी संख्या में प्रजातियों को एक ही स्थान या एक ही मौसम में लगा पाना सम्भव नहीं होता है। अतः यह वांछनीय है कि स्थान अथवा समयावधि के अन्तर्गत विचरण नियन्त्रण के साथ-साथ स्थान अथवा समयावधि के कारण होने वाले विचरण को भी नियंत्रित किया जाए। यह रिजॉल्वेबल ब्लॉक अभिकल्पनाओं के उपयोग द्वारा सम्भव हो सकता है। यहाँ पर, स्थान व समयावधि को पुनरावृत्ति के रूप में लिया जा सकता है तथा स्थान अथवा एक समयावधि के अन्तर्गत विचरण को ब्लॉकिंग द्वारा नियंत्रित किया जा सकता है। एक कृषि खेत परीक्षण में भूमि को पुनरावृत्ति आकार के अनुसार बड़े भूखण्डों में बाँटकर, प्रत्येक भूखण्ड को ब्लॉकों में उपविभाजित